FERMENTATION 7

When yeast is added to a solution of glucose ($C_6H_{12}O_6$), it can ferment the sugar solution to ethanol (C_2H_5OH) and carbon dioxide (CO_2). The ethanol remains in solution and the carbon dioxide is given off as a gas.

1 Write a balanced formula equation for the fermentation of glucose solution.

A student wanted to investigate the rate at which glucose was changed to ethanol by yeast. Below is part of his practical report.

Aim: To investigate the progress of a fermentation reaction by measuring mass changes

Method: 10.0 g of glucose was dissolved in 200 mL of water. A pinch of yeast (approx. 0.1 g) was added and the mixture was thoroughly shaken. The top of the conical flask was loosely plugged with cotton wool. (This was to allow carbon dioxide to escape, but prevent microbes from entering.) The flask and contents were placed on an electronic balance and data logger as shown in figure 1. The data logger was set to record the mass of the flask and contents each hour for 5 days.

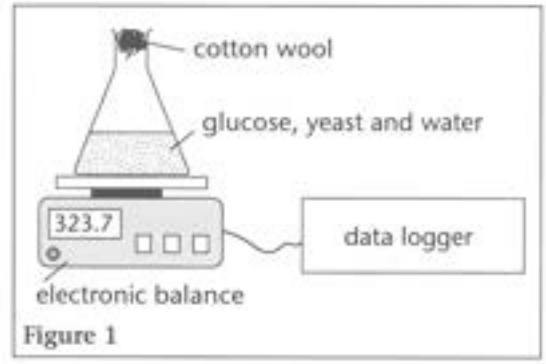


Figure 1

Results:

Mass of empty flask = 113.6 g

Mass of flask + water + glucose + yeast (at time 0) = 323.7 g

Sample readings:

Time (hours)	0	5	10	15	20	25	30	35	40	45	50	55	60
Mass (grams)	323.7	322.5	321.3	320.5	319.9	319.6	319.4	319.3	319.3	319.3	319.3	313.3	313.3

2 Explain why the mass of the flask and contents decreased over time.

3 Why did the student use a data logger rather than take the readings manually?

4 The sample readings do not show all of the readings. Why?

5 Draw up a table that shows time (hours) and total mass lost (grams).

6 Draw a graph to show the relationship between the mass lost (y-axis) and time (x-axis).

7 What is the dependent variable in the experiment?

8 What is the independent variable in the experiment?

9 When was the reaction complete? Explain your answer.

10 What mass of glucose was added?

11 What is the mass of one mole of glucose? Show your working. (Hint: C = 12.01, H = 1.008, O = 16.00)

12 How many moles of glucose were added? Show your working.

13 What mass of carbon dioxide evolved?

14 What is the mass of one mole of carbon dioxide?

15 How many moles of carbon dioxide were evolved? Show your working.

16 Assuming that one mole of carbon dioxide occupies 24 litres, what was the volume of carbon dioxide generated? Show your working.

17 After carrying out the experiment, the student wondered how changing the temperature would affect the rate of fermentation. Outline how he could carry out investigations to find out. Include details of how he could ensure that the results of the investigation were valid and reliable.

18 Define *fermentation*.

19 Give three examples of ways fermentation is used and what is produced.

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